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Daimler Center for Automotive IT Innovations

DCAITI-Project: Big Data Storage Solution for Mobility Data

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Overview

1. Recap
2. Plan
   2.1. 1st Phase - Research
   2.2. 2nd Phase - Installation and Benchmarking
3. Roadmap
4. References

Apache Accumulo [1]

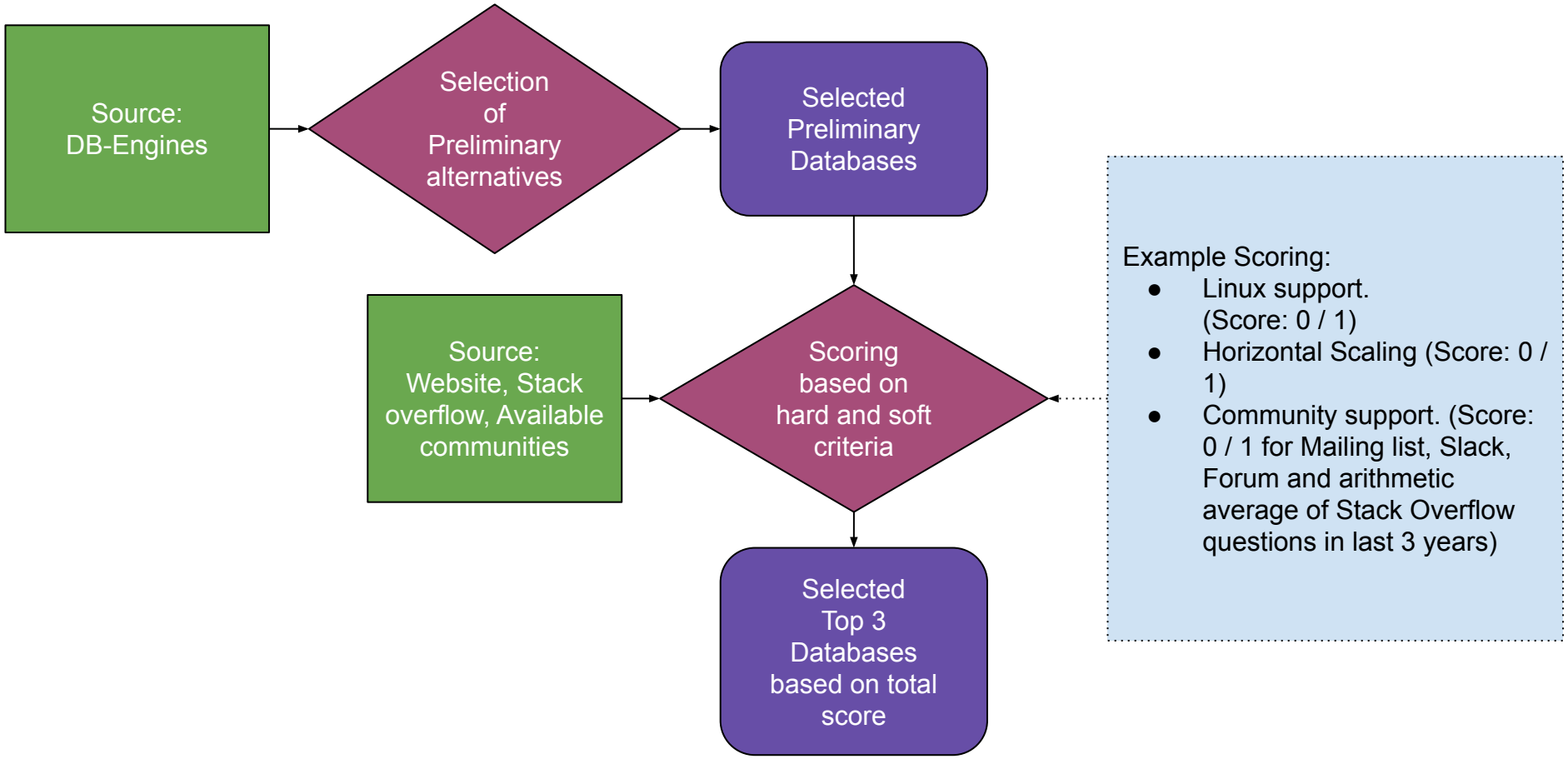
- NoSQL.
- Distributed.
- Wide Column.
- CP system (CAP).

Drawbacks

- Slow development pace.
- Low popularity.
- Hard to run algorithms on cluster.
- Complicated administration.

[2]

Research


Source:
DB-Engines
Selection
of
Preliminary
alternatives
Selected
Preliminary
Databases
Source:
Website, Stack
overflow, Available
communities
Scoring
based on
hard and soft
criteria
Selected
Top 3
Databases
based on total
score
Example Scoring:
Linux support.
(Score: 0 / 1)
Horizontal Scaling (Score: 0 / 1)
Community support. (Score: 0 / 1 for Mailing list, Slack, Forum and arithmetic average of Stack Overflow questions in last 3 years)

Plan - 1st Phase (Cont.)

Scoring

Hard Requirements

- Linux support on premise
- Linux Native Support
- Horizontal Scaling (Sharding)
- Horizontal Scaling (Replication)
- Active Development (2021-2023 stable releases)
- Community Support (Mailing List + Stack Overflow + Slack + Official Forum / Chat)
- Active Community (3 years' answered questions on Stack Overflow)
- Java API / Client / Driver
- Query Language

Soft Requirements

- CSV Support
- Parquet Support
- Security (Role Based Security)
- Geospatial Support
- Time Series Support

```
Scoring: 0 / 1 per requirement
```

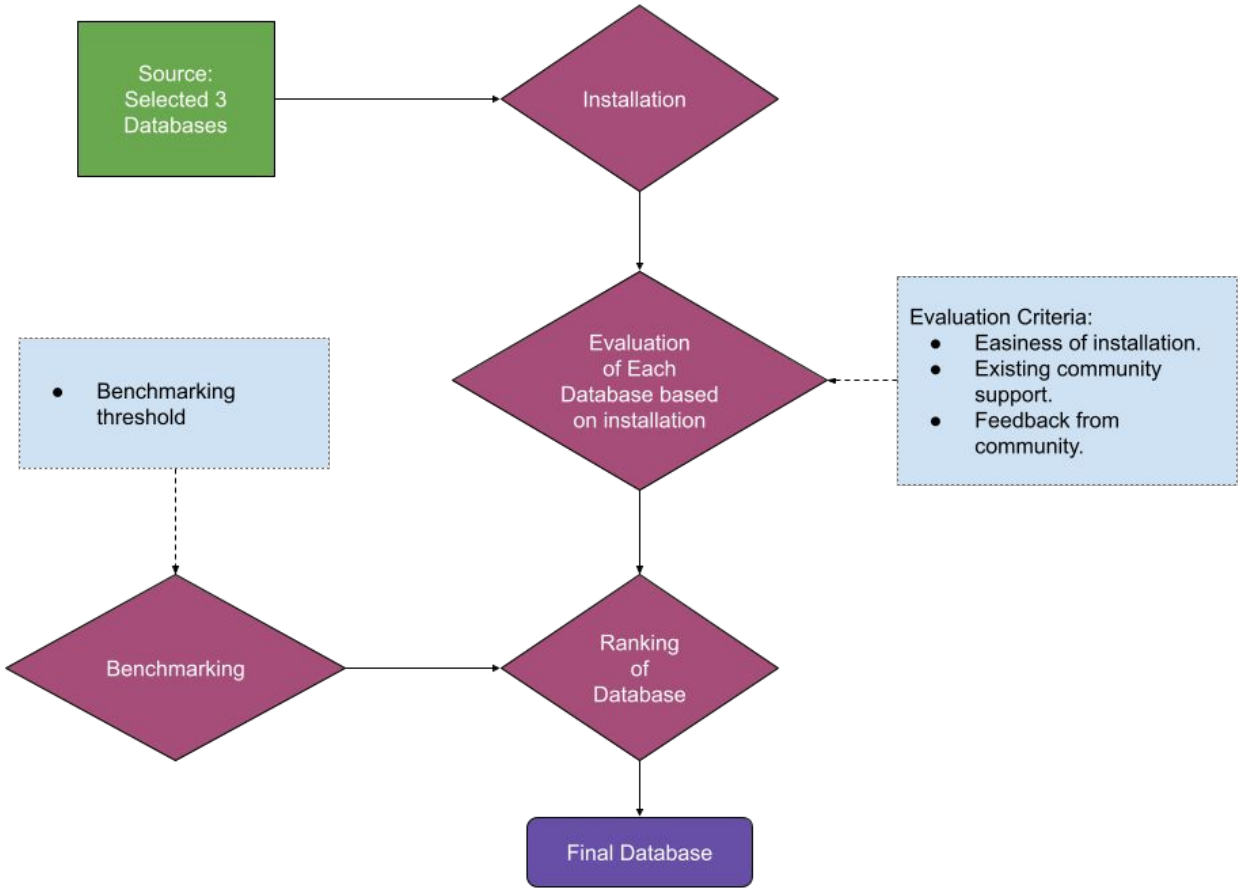
Plan - 1st Phase (Cont.)

PostgreSQL	8
MySQL	8
MariaDB	7.25
Redis	6.5
ScyllaDB	6.5
TimescaleDB	6.5
Apache Cassandra	5.75
Apache HBase	5.75
Google Cloud Bigtable	5
Azure Cosmos DB	5

[3]

```
Total Score = 0.75 * ∑ (Hard requirement score) + 0.25 * ∑ (Soft requirement score)
```

Installation and Benchmarking



Roadmap

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	April	May					June				July	
	CW17	CW18	CW19	CW20	CW21	CW22	CW23	CW24	CW25	CW26	CW27	CW28
Preparation for Sprint												
Finding Alternatives												
Installation												
Benchmarking												
Review + make presentation												

References

1. https://accumulo.apache.org
2. https://www.arresteddevops.com/sysadmins
3. https://github.com/mahmud2011/DCAITI-Projekt-Vertiefung-Vernetztes-und-Automatisiertes-Fahren/blob/main/docs/Primary%20Scoring.csv

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Thank You

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